

Renal cell carcinoma

Two forks decide everything — **clear cell vs non-clear cell**, and **IMDC risk**. Localized disease is surgical ± adjuvant pembrolizumab; metastatic clear cell gets an **IO backbone** (IO/TKI or ipi/nivo) with no predictive biomarker; **after IO, switch mechanism — never re-add IO**

Compiled from the Stanford Oncology Handbook 2025, GU: Renal Cell Carcinoma (Ruiz SM, Fan AC) · NCCN · primary trial reports and FDA actions verified against PubMed, September 2026.

WORKUP

- **Most are found incidentally** on imaging; the classic triad (flank pain, hematuria, mass) is seen in only ~10%
- **Paraneoplastic:** erythrocytosis (EPO), **hypercalcemia** (PTHrP), hypertension (renin), **Stauffer syndrome** (cholestasis without liver mets — resolves after nephrectomy)
- **CT abdomen/pelvis with contrast** (MRI if poor renal function, IVC thrombus, or to call fat in an angiomyolipoma) + **chest CT**. Brain MRI and bone scan when symptomatic or before systemic therapy in high-risk disease
- **Biopsy is not required before nephrectomy** for a solid enhancing mass — get it for small masses when it would change AS vs treatment, suspected lymphoma/metastasis/infection, or before systemic therapy without surgery
- Labs that feed IMDC: **CBC with differential, calcium**, plus LDH, creatinine, LFTs
- **Genetics referral (NCCN):** RCC at **age ≤46**, bilateral or multifocal tumors, ≥1 first/second-degree relative with RCC, or a somatic hit with germline implications

STAGING (AJCC 8TH)

T1	Confined to kidney, ≤7 cm — T1a ≤4 cm, T1b >4–7 cm
T2	Confined to kidney, >7 cm — T2a ≤10 cm, T2b >10 cm
T3	Major veins or perinephric fat, not beyond Gerota's — a renal vein / sinus or perirenal fat · b IVC below diaphragm · c IVC above diaphragm or caval wall
T4	Beyond Gerota's fascia, incl. contiguous ipsilateral adrenal
Stage	I T1N0 · II T2N0 · III T1–2N1 or T3 any N · IV T4 or M1

Grade (WHO/ISUP) is **nucleolar**: G1–G3 by nucleolar prominence at 400×/100×; G4 = marked **pleomorphism, giant cells, rhabdoid** or **sarcomatoid**.

IMDC RISK — THE 6 FACTORS

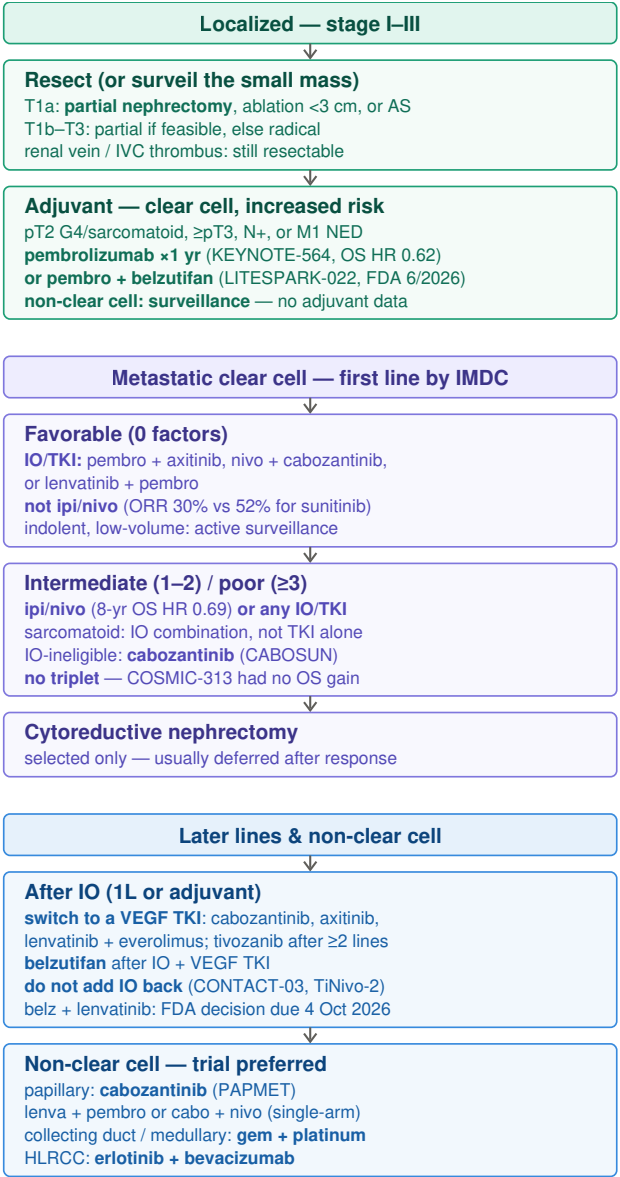
KPS <80%	Hemoglobin < LLN
Diagnosis → systemic tx <1 year	Calcium > ULN (corrected)
Neutrophils > ULN	Platelets > ULN

Favorable = 0 · Intermediate = 1–2 · Poor = ≥3. Two clinical + four labs. Prognostic, and it is how the 1L trials were stratified — but it does not predict IO vs TKI benefit except that ipi/nivo underperforms in favorable risk.

AFTER NEPHRECTOMY

- **Surveillance** (NCCN, risk-adapted): history, labs and chest/abdominal imaging every 3–6 months for ~3 years after stage II–III, then annually to year 5 and beyond — **RCC recurs late**, sometimes a decade or more after surgery
- **Unusual metastatic sites are classic:** pancreas, thyroid, adrenal, skin, and hypervascular brain lesions that bleed

TREATMENT PATHWAY



HISTOLOGY

Subtype	Biology · what it means
Clear cell ~75%	VHL loss (3p) → HIF-2α accumulation → VEGF . Why VEGF TKIs, IO and belzutifan work; chemo-resistant
Papillary ~10–15%	MET-driven subset (chr 7 gain, MET alterations). Cabozantinib best randomized data
Chromophobe ~5%	Usually indolent; IO barely works . BHD (FLCN) association
Collecting duct <1%	Aggressive, often metastatic at diagnosis — treat like urothelial: gemcitabine + platinum
Renal medullary <1%	Young patients with sickle cell trait ; SMARCB1 (IN11) loss. Platinum-based chemo; dismal prognosis
FH-deficient	HLRCC-associated (formerly 'type 2 papillary'); metastasizes when small
Translocation	TFE3/TFEB (MiT family); children and young adults
Sarcomatoid <i>any subtype</i>	Not a subtype but dedifferentiation — poor prognosis, yet the best responders to IO

LOCALIZED DISEASE & SURGERY

- **Small renal mass (≤4 cm): active surveillance is legitimate**, especially in the elderly or comorbid — many are benign or indolent. Serial imaging; intervene for growth (~>5 mm/yr) or size >3–4 cm. **Ablation** (cryo/RFA) for <3 cm
- **Partial nephrectomy preferred for T1a** (and T1b when feasible) to protect renal function. The one RCT, **EORTC 30904**, did not show OS non-inferiority (10-yr OS 75.7% vs 81.1%) but was underpowered — nephron-sparing remains standard
- **IVC tumor thrombus is not a contraindication** to resection. Lymphadenectomy has no proven OS benefit; no role for adjuvant RT
- **Cytoreductive nephrectomy is no longer routine:** CARMENA — sunitinib alone non-inferior (OS 18.4 vs 13.9 mo); SURTIME — deferred CN after 3 cycles of sunitinib gave longer OS (exploratory). Keep it for good PS, low-volume disease, symptomatic primary, or a responder with residual primary
- **Metastasectomy / SBRT** for oligometastatic disease, especially with a long disease-free interval

PEARLS

- **No chemotherapy for clear cell** — the only chemo-sensitive RCCs are collecting duct and medullary
- **Renal mass + sickle cell trait in a young Black patient = medullary carcinoma** until proven otherwise
- **Belzutifan is the VHL drug** (hereditary) **and** a later-line drug in sporadic ccRCC — same target, HIF-2α
- **Cabozantinib appears at every step:** 1L (with nivo or alone), 2L after IO (METEOR, CONTACT-03 control arm), and papillary (PAPMET)

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ADJUVANT — WHY KEYNOTE-564 STOOD ALONE

Trial	Agent · result
KEYNOTE-564	Pembrolizumab ×1 yr vs placebo — DFS HR 0.68; OS HR 0.62 (48-mo OS 91.2% vs 86.0%). The first adjuvant therapy in RCC with an OS benefit
LITESPARK-022	Pembro + belzutifan vs pembro — DFS HR 0.72, 24-mo DFS 80.7% vs 73.7%. OS immature (HR 0.78, NS); grade ≥3 AEs 52% vs 30% . FDA approved 12 Jun 2026
RAMPART <i>abstract only</i>	Durvalumab + tremelimumab vs monitoring — DFS HR 0.65; durvalumab alone NS (HR 0.74). Not approved
CheckMate 914 A	Nivo + ipi ×24 wk — DFS HR 0.92, negative ; 32% stopped for toxicity, 4 treatment deaths
CheckMate 914 B	Nivolumab alone — DFS HR 0.87, negative
PROSPER	One neoadjuvant nivo dose + 9 adjuvant — RFS HR 0.94, stopped for futility (included non-clear cell)
IMmotion010	Atezolizumab ×1 yr — DFS HR 0.93, negative
S-TRAC	Sunitinib ×1 yr — DFS 6.8 vs 5.6 yr (HR 0.76), no OS benefit (HR 0.92). FDA-approved 2017, rarely used
ASSURE · PROTECT · ATLAS	Sunitinib or sorafenib (DFS HR 1.02 / 0.97) · pazopanib (HR 0.86) · axitinib (HR 0.87, stopped for futility) — all negative

Candidate explanations for the IO split — anti-PD-1 vs anti-PD-L1, inclusion of resected M1 NED patients, adding CTLA-4 toxicity, PROSPER's single neoadjuvant dose — **none is proven**. Adjuvant therapy is for clear cell only.

FIRST-LINE CLEAR CELL — ALL VS SUNITINIB UNLESS STATED

Regimen · trial	Long-term result
Ipi + nivo <i>CheckMate 214</i>	Int/poor : OS HR 0.69 at 8 yr, CR 11.8% vs 2.6%, median DOR 82.8 mo — the most durable tail. Favorable : OS HR 0.82 (NS), ORR 30% vs 52%. PFS co-primary missed its α
Pembro + axitinib <i>KEYNOTE-426</i>	Final (43 mo): OS HR 0.73 , PFS HR 0.68, ORR 60% vs 40%. All risk groups
Nivo + cabozantinib <i>CheckMate 9ER</i>	Final (5.6 yr): OS 46.5 vs 35.5 mo, HR 0.79 ; PFS 16.4 vs 8.3 mo; ORR 55.7% vs 27.4%, CR 13.9%. Best QoL data
Lenvatinib + pembro <i>CLEAR</i>	PFS 23.9 vs 9.2 mo (HR 0.47), ORR 71.3% — highest response rate. Final OS HR 0.79, medians equal (53.7 vs 54.3 mo)
Cabo + nivo + ipi <i>COSMIC-313</i>	vs ipi/nivo, int/poor — PFS 16.6 vs 11.2 mo (HR 0.82) but OS HR 1.02 , grade 3–4 AEs 79% vs 56%. Triplet rejected
Cabozantinib <i>CABOSUN</i>	Int/poor, randomized phase 2 — PFS 8.2 vs 5.6 mo. The IO-ineligible option

No head-to-head trial between IO combinations and no predictive biomarker (PD-L1 does not select). Pick by risk group, comorbidity, and whether you want response (IO/TKI) or a durable treatment-free tail (ipi/nivo).

CHOOSING AMONG THE FIRST-LINE REGIMENS

- **Symptomatic, bulky, or need a response now** → IO/TKI; **lenvatinib + pembro** has the highest ORR and longest PFS
- **Int/poor risk, want a chance at a durable treatment-free remission** → **ipi/nivo**; also avoids chronic TKI toxicity. Expect front-loaded irAEs
- **Favorable risk** → IO/TKI (ipi/nivo lost on ORR and PFS); indolent low-volume disease can be watched
- **Autoimmune disease, transplant, or IO-ineligible** → cabozantinib (int/poor) or sunitinib/pazopanib
- **Relapse after adjuvant pembrolizumab**: no randomized data — most use a VEGF TKI-based regimen, especially if relapse occurs on or soon after IO

SECOND LINE AND BEYOND

Trial	Setting · result
LITESPARK-011	Belzutifan + lenvatinib vs cabozantinib after anti-PD-(L)1 — PFS 14.8 vs 10.7 mo (HR 0.70), ORR ~53% vs ~40%; interim OS HR 0.85, not significant (final pending). FDA action date 4 Oct 2026
LITESPARK-005	Belzutifan vs everolimus after IO + TKI — ORR 21.9% vs 3.5%, 18-mo PFS 24% vs 8%, but median PFS 5.6 mo in both and OS HR 0.88, NS . FDA 14 Dec 2023
METEOR	Cabozantinib vs everolimus after VEGF TKI — OS 21.4 vs 16.5 mo (HR 0.66), PFS HR 0.51
CheckMate 025	Nivolumab vs everolimus after VEGF TKI — OS 25.0 vs 19.6 mo (HR 0.73); the IO-naïve 2L option
Study 205	Lenvatinib + everolimus vs everolimus, randomized phase 2 — PFS 14.6 vs 5.5 mo (HR 0.40)
TIVO-3	Tivozanib vs sorafenib after ≥2 lines — PFS 5.6 vs 3.9 mo; no OS benefit
CONTACT-03	Atezolizumab + cabo vs cabo after IO — PFS HR 1.03, OS HR 0.94: negative
TiNivo-2	Tivozanib + nivolumab vs tivozanib after IO — PFS 5.7 vs 7.4 mo, negative

IO rechallenge **does not work** — two negative phase 3s. After IO, the next drug is a VEGF TKI; belzutifan comes after IO + TKI.

NON-CLEAR CELL — MOSTLY SINGLE-ARM DATA

Histology / trial	Evidence
Papillary <i>SWOG S1500 (PAPMET)</i>	Cabozantinib vs sunitinib , randomized phase 2 — PFS 9.0 vs 5.6 mo (HR 0.60), ORR 23% vs 4%; OS NS. Savolitinib and crizotinib arms closed for futility
Mixed nccRCC <i>KEYNOTE-B61</i>	Lenvatinib + pembro , single-arm — ORR 49% (51% at 2 yr), sarcomatoid 47%
Mixed nccRCC <i>phase 2</i>	Cabo + nivo — ORR 47.5% papillary/unclassified; 0 of 7 chromophobe responded
Mixed nccRCC <i>KEYNOTE-427 B</i>	Pembrolizumab alone — ORR 26.7%, mPFS 4.2 mo
Chromophobe	IO largely inactive; VEGF TKI ± mTOR (e.g., lenvatinib + everolimus). Trial preferred
Collecting duct · medullary	Gemcitabine + cisplatin/carboplatin ; TKIs and IO rarely help. Check sickle status in a young patient with a renal mass
Sarcomatoid	Use an IO combination whatever the parent histology — the highest CR rates in 1L subgroups

HEREDITARY SYNDROMES (ALL AUTOSOMAL DOMINANT)

Syndrome / gene	Renal tumor · other features · management
VHL <i>VHL, 3p</i>	Clear cell, multifocal/bilateral ; CNS/retinal hemangioblastomas, pheochromocytoma, pNET, pancreatic cysts. Belzutifan (LITESPARK-004: RCC ORR 49%; FDA Aug 2021) spares repeated surgery; resect when the largest tumor reaches ~3 cm
HLRCC <i>FH</i>	FH-deficient RCC — solitary, aggressive, metastasizes early: excise promptly, no surveillance . Cutaneous + uterine leiomyomas. Advanced: erlotinib + bevacizumab (ORR 72%, mPFS 21.1 mo)
Birt-Hogg-Dubé <i>FLCN</i>	Chromophobe / hybrid oncocytic , bilateral; fibrofolliculomas, lung cysts → pneumothorax
Tuberous sclerosis <i>TSC1/2</i>	Angiomyolipomas (rarely RCC); everolimus . Angiofibromas, SEGA, LAM
BAP1 TPDS <i>BAP1</i>	Aggressive clear cell ; uveal and cutaneous melanoma, mesothelioma
HPRC <i>MET</i>	Papillary , multifocal and bilateral
Hereditary PGL/ PCC <i>SDHB/C/D</i>	SDH-deficient RCC ; paraganglioma, pheochromocytoma, GIST

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REGIMEN DOSING

Regimen	Dosing
Adjuvant pembrolizumab <i>KEYNOTE-564</i>	200 mg IV q3wk ×17 or 400 mg q6wk (~1 yr)
+ Belzutifan <i>LITESPARK-022</i>	Belzutifan 120 mg PO daily ×up to 54 wk with pembrolizumab 200 q3wk / 400 q6wk ×12 mo
Pembro + axitinib	Axitinib 5 mg PO BID (titrate to 7 then 10 mg BID if BP controlled) + pembrolizumab 200 mg q3wk / 400 q6wk, IO up to 2 yr
Ipi + nivo	Nivolumab 3 mg/kg + ipilimumab 1 mg/kg IV q3wk ×4 → nivolumab 240 mg q2wk or 480 mg q4wk
Nivo + cabozantinib	Cabozantinib 40 mg PO daily (not 60) + nivolumab 240 q2wk / 480 q4wk, nivo max 2 yr
Lenvatinib + pembro	Lenvatinib 20 mg PO daily + pembrolizumab 200 q3wk / 400 q6wk, IO up to 2 yr
Cabozantinib	60 mg PO daily on an empty stomach; hold ≥3 wk before surgery
Belzutifan	120 mg PO daily — after IO + VEGF TKI, and for VHL tumors
Belzutifan + lenvatinib	120 mg + 20 mg daily (LITESPARK-011; pending FDA)
Tivozanib	1.34 mg PO daily, 21 days on / 7 off — after ≥2 prior systemic lines
Lenvatinib + everolimus	Lenvatinib 18 mg + everolimus 5 mg PO daily
Axitinib alone	5 mg BID → 7 → 10 mg BID if BP ≤150/90 and tolerated
Sunitinib	50 mg PO daily, 4 wk on / 2 wk off (or 2 on / 1 off for tolerance)
Pazopanib	800 mg PO daily , empty stomach
Nivolumab alone	240 mg q2wk or 480 mg q4wk
Erlotinib + bevacizumab	Erlotinib 150 mg PO daily + bevacizumab 10 mg/kg IV q2wk — HLRCC / papillary

Pembrolizumab in RCC is 200 mg q3wk or 400 mg q6wk — not 300/600. Historical: high-dose IL-2 (ORR ~14%, rare durable CRs), temsirolimus (poor-risk, ARCC).

TOXICITY & MONITORING

- **Hypertension is the signature VEGF-TKI toxicity** — control BP before starting, home BP log, add agents early rather than stopping the TKI. Urine protein with lenvatinib and bevacizumab
- **Transaminitis on axitinib + pembrolizumab** (or cabo/lenva + IO): **hold both drugs**. TKI hepatitis resolves within days (short half-lives); **if ALT is not falling after ~1 week, or grade ≥3, treat as immune hepatitis** with prednisone 1–2 mg/kg. Restart the TKI first at reduced dose; rechallenge IO only if it wasn't the culprit
- **Pazopanib carries a boxed warning for hepatotoxicity** — LFTs before starting and every ~2 weeks through month 4; also QT prolongation
- **Hypothyroidism** from both TKIs (sunitinib, axitinib, lenvatinib) and IO — TSH every 4–6 wk on treatment
- **Belzutifan: on-target anemia** (HIF-2α drives EPO — decreased Hgb is the most common adverse reaction) and **hypoxia**: check Hgb and pulse oximetry at baseline and periodically; transfuse, hold or dose-reduce. **Boxed warning: embryo-fetal toxicity**, and it can make hormonal contraception ineffective
- **Ipi/nivo**: front-loaded irAEs — colitis, hepatitis, hypophyinitis, adrenal insufficiency. Check cortisol with fatigue or hyponatremia
- **Cabozantinib**: diarrhea, hand-foot, stomatitis, wound-healing and bleeding — dose reductions to 40/20 mg are routine
- **Sunitinib**: hand-foot, mucositis, cytopenias, cardiomyopathy. **mTOR inhibitors**: pneumonitis, hyperglycemia, hyperlipidemia
- **Bone metastases** (lytic, ~1/3): zoledronic acid (renally dosed) or denosumab; **brain metastases** — SRS before or alongside systemic therapy

SPECIAL SITUATIONS

- **Brain metastases**: treat locally first (SRS or resection); patients with treated, stable brain mets can receive standard systemic therapy
- **Dialysis / poor renal function**: IO needs no renal adjustment; dose-adjust zoledronic acid, and **denosumab carries a severe-hypocalcemia risk in advanced CKD**
- **Pre-existing autoimmune disease or solid-organ transplant**: avoid IO; TKI monotherapy
- **Surgery on a TKI**: hold cabozantinib ≥3 wk before and resume only after wound healing; bevacizumab ≥28 days

KEY TRIALS

KEYNOTE-564	Adjuvant pembrolizumab — DFS HR 0.68, OS HR 0.62
LITESPARK-022	Adding belzutifan to adjuvant pembro — DFS HR 0.72, OS immature
CheckMate 914 · PROSPER · IMmotion010	Adjuvant nivo ± ipi, perioperative nivo, atezolizumab — all negative
CheckMate 214	Ipi/nivo, int/poor — 8-yr OS HR 0.69, durable CRs
CLEAR	Lenva + pembro — PFS 23.9 vs 9.2 mo, ORR 71%
COSMIC-313	Triplet cabo + ipi/nivo — PFS gain, no OS gain
CARMENA · SURTIME	Immediate cytoreductive nephrectomy not needed; defer and select
METEOR	Cabozantinib after TKI — OS HR 0.66
LITESPARK-005 / -011	Belzutifan after IO + TKI (ORR 22%); + lenvatinib post-IO, PFS HR 0.70
CONTACT-03 · TINIvo-2	IO rechallenge after IO — no benefit
PAPMET	Papillary: cabozantinib beats sunitinib on PFS
LITESPARK-004	Belzutifan in VHL disease — RCC ORR 49%
S-TRAC · ASSURE	Adjuvant sunitinib: DFS+ but no OS (S-TRAC); negative (ASSURE)
KEYNOTE-426 · 9ER	IO/TKI vs sunitinib — final OS HR 0.73 / 0.79
CheckMate 025	Nivolumab after TKI — OS 25.0 vs 19.6 mo
NCI bev + erlotinib	HLRCC papillary RCC — ORR 72%, mPFS 21.1 mo
EORTC 30904	Partial vs radical nephrectomy — the only RCT; underpowered